



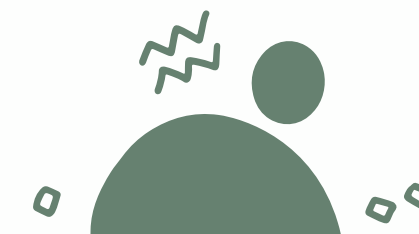
SMART CITY IOT SYSTEM

PATRICIA PARUNGAO

MARIA NICOLE JAURIGUE

BERNADETTE OGMA

PATRICK CANETE



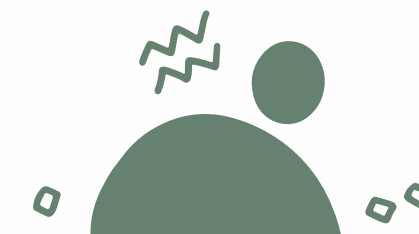


WHAT IS A SMART CITY IOT SYSTEM?

A network of interconnected devices, sensors, and platforms that optimize urban services including:

- Traffic management (smart traffic lights, parking sensors)
- Public safety (surveillance cameras, emergency response systems)
- Utilities (smart grids, water quality monitors)
- Waste management (fill-level sensors)

Core Goal: Improve quality of life, reduce resource waste, and enhance operational efficiency for residents and local governments.





IDENTIFIED RISKS

1. CYBERSECURITY BREACHES

- Vulnerabilities in unpatched sensors/devices can be exploited to gain unauthorized access
- Risks include data theft (residential/location data) and system manipulation (e.g., traffic light hijacking)



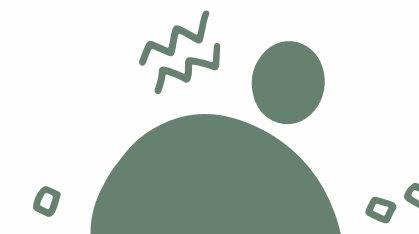
IDENTIFIED RISKS

2. DATA PRIVACY VIOLATIONS

- Mass collection of real-time data (location, movement patterns, energy usage) creates privacy risks
- Potential for misuse by third parties or unauthorized internal access

3. SYSTEM RELIABILITY FAILURES

- Single points of failure can disrupt critical services (e.g., power grid outages from sensor malfunctions)
- Natural disasters or physical tampering with devices can cause widespread operational issues





SOLUTION FOR CYBERSECURITY BREACHES

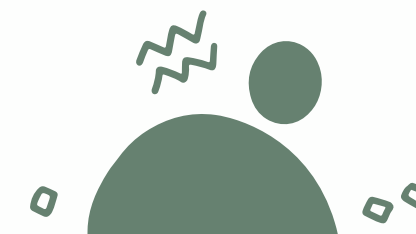
Proposed Measures:

1. Mandatory Device Security Standards

- Require all IoT devices to meet national cybersecurity certifications (e.g., ISO/IEC 27040) before deployment



Implement automatic over-the-air (OTA) patch management for all connected systems





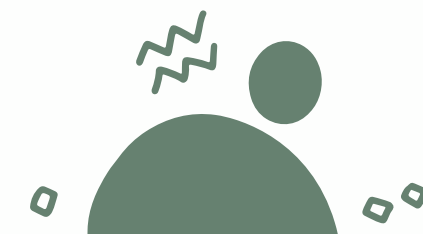
SOLUTION FOR CYBERSECURITY BREACHES

2. Network Segmentation

- Isolate critical systems (e.g., power grid, emergency services) from non-critical ones (e.g., waste management sensors)
- Use firewalls and virtual private networks (VPNs) to limit lateral movement in case of a breach

3. Continuous Monitoring

- Deploy AI-powered threat detection tools to identify unusual activity patterns
- Establish a 24/7 security operations center (SOC) to respond to incidents immediately





STRUCTURED DATA PRIVACY GOVERNANCE

SOLUTION FOR DATA PRIVACY VIOLATIONS

Proposed Measures:

1. Privacy-by-Design Implementation

- Collect only necessary data (minimization principle) and anonymize datasets where possible
- Use differential privacy techniques to protect individual identities in aggregated analytics



STRUCTURED DATA PRIVACY GOVERNANCE

2. Transparent Data Governance Policies

- Publish clear public notices on what data is collected, how it is used, and retention periods
- Establish a dedicated data protection office (DPO) to handle resident inquiries and complaints

3. User Control Mechanisms

- Provide residents with digital portals to access, correct, or delete their data
- Offer opt-out options for non-critical data collection (e.g., traffic flow analytics)



IMPROVED SYSTEM RELIABILITY

SOLUTION FOR SYSTEM RELIABILITY FAILURES

Proposed Measures:

1. Redundant Infrastructure Design

- Eliminate single points of failure by building backup systems for critical services
- Use distributed cloud architecture to ensure service continuity during local outages



IMPROVED SYSTEM RELIABILITY

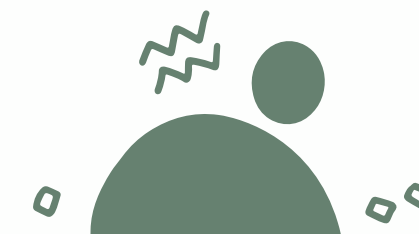
SOLUTION FOR SYSTEM RELIABILITY FAILURES

2. Physical and Environmental Protection

- Encase outdoor sensors in tamper-proof, weather-resistant housing
- Install motion detectors and surveillance at key equipment locations to prevent physical attacks

3. Predictive Maintenance Programs

- Use sensor data to monitor device health and predict failures before they occur
- Schedule regular on-site inspections for high-priority systems (e.g., flood monitoring sensors)





CONCLUSION

- Smart City IoT systems offer transformative benefits for urban communities, but success depends on addressing key risks:
- Cybersecurity breaches → Mitigate with standardized security and monitoring
- Data privacy violations → Govern with transparency and user control
- System reliability failures → Strengthen with redundancy and predictive care



THANK YOU

